Gamma and Beta functions:

Equivalent definitions of Gamma function

$$\Gamma(x) = \lim_{n\to\infty} \frac{n!\, n^x}{x(x+1)(x+2)\cdots(x+n)}$$

$$\Gamma(x) = \int_0^\infty dt\, e^{-t}\, t^{x-1} \qquad x>0$$

usful properties:

$$\Gamma(x) = \frac{\Gamma(x+k)}{x(x+1)(x+2)\cdots(x+k-1)} \qquad k=1,2,\ldots \quad x>-k$$

$$\Gamma(1)=1\,, \quad \Gamma(1/2) = \sqrt{\pi}$$

usuful substitutions $t=u^2$

$$\Gamma(x) = 2\int_0^\infty e^{-u^2} u^{2x-1}\, du \qquad x>0$$

$t=\ln(1/u)$

$$\Gamma(x) = \int_0^1 \left(\ln\frac{1}{u}\right)^{x-1} du \qquad x>0$$

$$\int_0^{\pi/2} \cos^{2x-1}\theta\, \sin^{2y-1}\theta\, d\theta = \frac{\Gamma(x)\Gamma(y)}{2\Gamma(x+y)}$$

$x, y > 0$

Legendre duplication formula

$$2^{2x-1}\,\Gamma(x)\,\Gamma(x+1/2) = \sqrt{\pi}\,\Gamma(2x)$$

Weierstrass infinite product

$$\frac{1}{\Gamma(x)} = x e^{\gamma x} \prod_{n=1}^{\infty} \left(1+\frac{x}{n}\right) e^{-x/n}$$

$$\Rightarrow \quad \Gamma(x)\Gamma(-x) = -\frac{\pi}{x \sin \pi x}$$

$$\Rightarrow \quad \Gamma(x)\,\Gamma(1-x) = \frac{\pi}{\sin \pi x}$$

Beta function useful formulas.

$$B(x,y) = \int_0^1 t^{x-1}\,(1-t)^{y-1}\,dt \qquad x>0,\ y>0$$

$$t = \frac{u}{1+u}$$

$$= \int_0^{\infty} \frac{u^{x-1}}{(1+u)^{x+y}}\,du = \frac{\Gamma(x)\,\Gamma(y)}{\Gamma(x+y)}$$

Claim: $\Gamma(x) = (\ln b)^x \int_0^\infty t^{x-1} b^{-t}\, dt \qquad x>0,\ b>0$

proof:

$$\Gamma(x) = \int_0^\infty dt\ e^{-t}\ t^{x-1}$$

Since $b = e^{\ln b}$, $\quad t = u \ln b \qquad b>0$

$dt = du \ln b$

$$\Gamma(x) = \int_0^\infty \ln b\, du\ e^{-u\ln b}\ (u \ln b)^{x-1}$$

$$= (\ln b)^x \int_0^\infty du\ u^{x-1}\ b^{-u}$$

Claim: $\int_a^\infty e^{2ax - x^2} dx = \frac{1}{2}\sqrt{\pi}\ e^{a^2}$

proof: $\int_a^\infty e^{2ax-x^2} dx = \int_a^\infty e^{-(x^2 - 2ax + a^2 - a^2)} dx$

$$= e^{a^2} \int_a^\infty e^{-(x-a)^2} dx \qquad x \longrightarrow x+a$$

$$= e^{a^2} \int_0^\infty e^{-x^2} dx \qquad u = x^2,\ du = 2x\,dx$$

$$= \frac{1}{2} e^{a^2} \int_0^\infty u^{-1/2} e^{-u}\, du$$

$$= \frac{1}{2} e^{a^2} \Gamma(1/2)$$

Claim: $\int_0^1 x^k (\ln x)^n \, dx = \frac{(-1)^n \, n!}{(k+1)^{n+1}}$ $\quad k > -1$, $n = 0, 1, 2, \ldots$

proof: $x = e^{-t} \qquad dx = -e^{-t} dt$

$$\int_0^\infty (-t)^n \, e^{-kt} \, e^{-t} \, dt$$

$$= (-1)^n \int_0^\infty t^n \, e^{-(k+1)t} \, dt$$

$$\frac{(-1)^n}{(k+1)^{n+1}} \int_0^\infty dt \; t^n \, e^{-t} = \frac{(-1)^n}{(k+1)^{n+1}} \Gamma(n+1)$$

claim: $B(x+1, y) + B(x, y+1) = B(x, y) \qquad x, y > 0$

proof :

$$\int_0^\infty \left[\frac{u^x}{(1+u)^{x+y+1}} + \frac{u^{x-1}}{(1+u)^{x+y+1}} \right] du$$

$$= \int_0^\infty \frac{u^x + u^{x-1}}{(1+u)^{x+y+1}} \, du = \int_0^\infty \frac{(u+1)\, u^{x-1}}{(1+u)^{x+y+1}} \, du$$

$$= \int_0^\infty \frac{u^{x-1}}{(1+u)^{x+y}} \, du = B(x, y)$$

claim: $\Gamma(x) = \int_0^\infty e^{-t}(t-x)\,t^{x-1}\ln t\,dt$

proof: $\int_0^\infty e^{-t}(t-x)\,t^{x-1}\ln(t)\,dt$

$= \int_0^\infty \ln(t)\,e^{-t}\,t^{x}\,dt - x\int_0^\infty \ln(t)\,e^{-t}\,t^{x-1}\,dt$

$= \int_0^\infty e^{-t}\frac{d}{dx}(t^x)\,dt - x\int_0^\infty e^{-t}\frac{d}{dx}(t^{x-1})\,dt$

$= \frac{d}{dx}\Gamma(x+1) - x\frac{d}{dx}\Gamma(x)$

$= \Gamma'(x+1) - x\Gamma'(x) = \Gamma'(x+1) - x\frac{d}{dx}\left(\frac{\Gamma(x+1)}{x}\right)$

$= \Gamma'(x+1) - \Gamma'(x+1) + x\frac{\Gamma(x+1)}{x^2}$

$= \frac{\Gamma(x+1)}{x} = \Gamma(x)$

Claim: $\Gamma(x)\cos ax = b^x \int_0^\infty t^{x-1} e^{-bt\cos a} \cos(bt\sin a)\,dt$

proof: $b^x \int_0^\infty t^{x-1} e^{-b+\cos a} \cos(bt\sin a)\,dt$

$$= \int_0^\infty t^{x-1} e^{-t\cos a} \cos(t\sin a)\,dt$$

$$\cos(t\sin a) = \mathrm{Re}\{ e^{it\sin a} \}$$

$$= \mathrm{Re}\left\{ \int_0^\infty t^{x-1} e^{-t(\cos a - i\sin a)}\,dt \right.$$

$$= \mathrm{Re}\left\{ \int_0^\infty t^{x-1} e^{-e^{-ia}t}\,dt \right\}$$

$u = e^{-ia}t$

$du = e^{-ia}dt$

$$= \mathrm{Re}\left\{ \int_0^\infty (e^{ia}a)^{x-1} e^{-u} e^{ia}\,du \right\}$$

$$= \mathrm{Re}\{ e^{iax} \} \int_0^\infty u^{x-1} e^{-u}\,du$$

$$= \cos ax\, \Gamma(x)$$

Claim: $B(x, y+1) = \frac{y}{x} B(x+1, y) = \frac{y}{x+y} B(x, y)$

$$B(x, y+1) = \frac{\Gamma(x)\,\Gamma(y+1)}{\Gamma(x+y+1)} = \frac{\Gamma(x+1)}{x} \frac{y\Gamma(y)}{\Gamma(x+y+1)}$$

$$= \frac{y}{x} \frac{\Gamma(x+1)\Gamma(y)}{\Gamma(x+y+1)} = \boxed{\frac{y}{x} B(x+1, y)}$$

$$= \frac{\Gamma(x)\, y\Gamma(y)}{(x+y)\,\Gamma(x+y)} = \frac{y}{x+y} B(x, y).$$

4) $\int_a^b (b-x)^{m-1} (x-a)^{n-1} dx$

$b \to 1$
$a \to 0$

$x = (b-a)t + a$ Same as above.

$\downarrow \quad x = (b-a)dt$

$$b - x = b - (b-a)t - a$$
$$= (b-a)(1-t)$$
$$x - a = (b-a)t + \cancel{a} - \cancel{a}$$

$$\Rightarrow \int_0^1 (b-a)^{m-1} (1-t)^{m-1} t^{n-1} (b-a)^{n-1} (b-a) dt$$

$$= (b-a)^{n+m-1} B(n, m)$$

Claim : $\int_0^\infty \frac{x^{p-1}\ln(x)}{1+x}\,dx = -\pi^2 \csc p\pi \cot p\pi$

proof $\int_0^\infty \frac{x^{p-1}\ln(x)}{1+x}\,dx = \int_0^\infty \frac{d}{dp}\left[\frac{x^{p-1}}{1+x}\right]dx$

$= \frac{d}{dp}\left[\int_0^\infty dx \frac{x^{p-1}}{1+x}\right] = \frac{d}{dp}\left(B(p,1-p)\right)$

$= \frac{d}{dp}\left(\pi^2 \csc p\pi\right) = -\pi^2 \csc p\pi \cot p\pi$

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claim: $B(x,y) = \int_0^1 \frac{t^{x-1}+t^{y-1}}{(t+1)^{x+y}}\,dt$

proof: $1 \to \pi/2 \qquad 0 \to 0$

$t = \tan^2\left(\frac{u}{2}\right) = \tan^2\left(\frac{u}{2}\right) \qquad dt = \tan\left(\frac{u}{2}\right)\sec^2\left(\frac{u}{2}\right)du$

$\int_0^{\pi/2} \frac{\tan^{2x-2} + \tan^{2y-2}}{(\sec^2)^{x+y}} \tan\left(\frac{u}{2}\right)\sec^2\left(\frac{u}{2}\right)du$

$= \int_0^{\pi/2} \frac{\tan^{2x-1} + \tan^{2y-1}}{(\sec^2)^{x+y-1}}\,du$

$$\int_0^{\pi/2} \frac{\tan^{2x-1}}{\sec^{2x+2y-2}} du = \int_0^{\pi/2} \frac{\sin^{2x-1}\left(\frac{u}{2}\right)}{\cos^{2x-1-2x-2y+2}\left(\frac{u}{2}\right)} du$$

$$= \int_0^{\pi/2} \sin^{2x-1}\left(\frac{u}{2}\right) \cos^{2y-1}\left(\frac{u}{2}\right) du$$

$$= 2\int_0^{\pi/4} \sin^{2x-1}(u) \cos^{2y-1}(u)\, du$$

$$\Rightarrow 2\left[\int_0^{\pi/4} \sin^{2x-1} u \cos^{2y-1} u\, du + \int_0^{\pi/4} \sin^{2y-1} u \cos^{2x-1} u\, du\right]$$